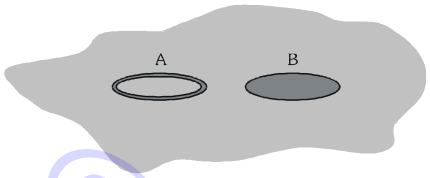
SECTION-I

Single Correct Answer Type **7 Q. [3 M (–1)]**

1. A rigid ring A and a rigid thin disk B both made of same material, when gently placed on water, just manage to float due to surface tension as shown in the figure. Both the ring and the disk have same radius. What can you conclude about their masses?

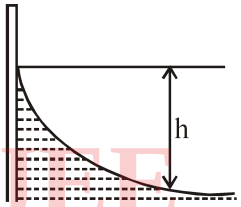


(A) Both have the same mass.
(B) Mass of the ring is half of that of the disk.
(C) Mass of the ring is double to that of the disk.
(D) More information is needed to decide.

2. Two soap bubbles with radii r_1 and r_2 ($r_1 > r_2$) come in contact. Their common surface has a radius of curvature r.

(A) $r = \frac{r_1 + r_2}{2}$ (B) $r = \frac{r_1 \ r_2}{r_1 - r_2}$ (C) $r = \frac{r_1 \ r_2}{r_1 + r_2}$ (D) $r = \sqrt{r_1 \ r_2}$

3. Water of density ρ in a clean aquarium forms a meniscus, as illustrated in the figure. Calculate the difference in height h between the centre and the edge of the meniscus. The surface tension of water is γ.



(A) $\sqrt{\frac{2\gamma}{\rho g}}$ (B) $\sqrt{\frac{\gamma}{\rho g}}$

(C) $\frac{1}{2}\sqrt{\frac{\gamma}{\rho g}}$ (D) $2\sqrt{\frac{\gamma}{\rho g}}$

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4. Figure (a) and (b) show water drop and mercury drop in two identical conical glass pipes, then

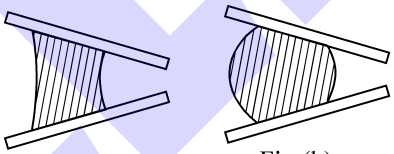
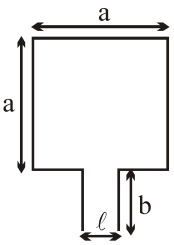


Fig.(a) Fig.(b)

(A) The water drop tends to move towards narrow end and mercury drop towards wide end.
(B) The water drop tends to move towards wide end and mercury drop towards narrow end.
(C) both tend to move towards wide end.
(D) both tend to move towards narrow end.

5. A light open rigid wire frame floats on the surface of water as shown in figure. What force will act on the frame, immediately after some soap solution is dropped inside it? α_1 and α_2 are the surface tensions of water and soap respectively ($\alpha_1 > \alpha_2$)



(A) zero (B) $(\alpha_1 - \alpha_2)\ \ell$
(C) $(\alpha_1 + \alpha_2)(4a + 2b + \ell)$ (D) $(\alpha_1 - \alpha_2)(4a + 2b - \ell)$

6. If a drop of liquid breaks into smaller droplets, it results in lowering of temperature of the droplets. Let a drop of radius *R*, break into *N* small droplets each of radius *r*. Estimate the change in temperature.

(A) $\frac{2T}{\rho s}\left(\frac{1}{R} + \frac{1}{r}\right)$ (B) $\frac{4T}{\rho s}\left(\frac{1}{R} - \frac{1}{r}\right)$ (C) $\frac{3T}{\rho s}\left(\frac{1}{R} - \frac{1}{r}\right)$ (D) $\frac{2T}{\rho s}\left(\frac{1}{R} - \frac{1}{r}\right)$

7. A drop of water volume 0.05 cm^3 is pressed between two glass-plates, as a consequence of which, it spreads between the plates. The area of contact with each plate is 40 cm^2. If the surface tension of water is 70 dyne/cm, the minimum normal force required to seperate out the two glass plates in newton is approximately (assuming angle of contact is zero) :

(A) 45 N (B) 100 N (C) 90 N (D) None of these

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Multiple Correct Answer Type 6 Q. [4 M (–1)]

8. n drops of a liquid, each with surface energy E, joining to form a single drop

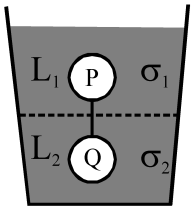
(A) some energy will be released in the process

(B) some energy will be absorbed in the process

(C) the energy released or absorbed will be $E(n-n^{2/3})$

(D) the energy released or absorbed will be $nE\,(2^{2/3}-1)$

9. Two spheres P and Q of equal radii have densities ρ_1 and ρ_2, respectively. The spheres are connected by a massless string and placed in liquids L_1 and L_2 of densities σ_1 and σ_2 and viscosities η_1 and η_2, respectively. They float in equilibrium with the sphere P in L_1 and sphere Q in L_2 and the string being taut (see figure). If sphere P alone in L_2 has terminal velocity $\vec{V}_P$ and Q alone in L_1 has terminal velocity $\vec{V}_Q$, then



(A) $\frac{|\vec{V}_P|}{|\vec{V}_Q|}=\frac{\eta_1}{\eta_2}$ (B) $\frac{|\vec{V}_P|}{|\vec{V}_Q|}=\frac{\eta_2}{\eta_1}$ (C) $\vec{V}_P.\vec{V}_Q>0$ (D) $\vec{V}_P.\vec{V}_Q<0$

10. A soap film (surface tension = T) is formed on a rectangular wire frame as shown in figure. A small, light, inextensible loop of thread of length "ℓ" is gently placed over the soap film. The loop stays on the film in the irregular fashion in which it was placed. Now, a hole is pricked in the film inside the loop with a needle and the thread finally comes to equilibrium position. Choose the correct option(s)

(A) The thread will take a square shape

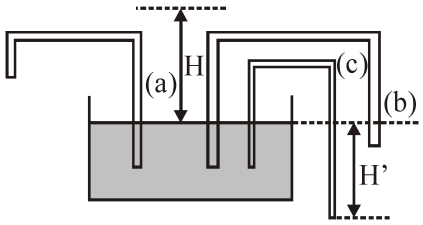
(B) The thread will take a circular shape



(C) Tension in the thread will be $2T\ell$

(D) Tension in the thread will be $\frac{T\ell}{\pi}$

11. Water which wets the walls of a vertical capillary tube rises to a height H within it. Three 'gallows', (a), (b) and (c), are made from the same tubing, and one end of each is placed into a large dish filled with water, as shown in the figure. (H' > H) then



(A) In case a water cannot flow out of the tube.

(B) In case b water cannot flow out of the tube.

(C) In case c water cannot flow out of the tube.

(D) In case c water will flow out of the tube.

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12. A sphere is dropped into a viscous liquid of viscosity η from some height. If the density of material and liquid are ρ and σ respectively ($\rho > \sigma$) then which of the following is incorrect.

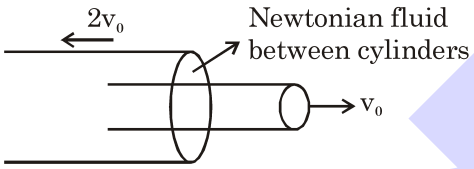
(A) the acceleration of the sphere just after entering the liquid is $g\left(\frac{\rho-\sigma}{\rho}\right)$

(B) Time taken to attain terminal speed $t \propto \rho^0$

(C) At terminal speed, the viscous force is maximum

(D) At terminal speed, the net force acting on the sphere is zero

13. Consider a situation in which there is a Newtonian fluid between two long coaxial uniformly moving cylinders of radius R and 2R.



(A) Force per unit length required to keep the inner cylinder moving is $4\pi\eta v_0$

(B) Force per unit length required to keep the inner cylinder moving is $\frac{6\pi\eta v_0}{\ell n 2}$.

(C) Distance of the point from the axis at which speed of fluid is zero is $2^{1/3}R$.

(D) Distance of the point from the axis at which speed of fluid is zero is $\sqrt{2}R$.

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Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Questions no. 14 to 16

When liquid medicine of density ρ is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop. We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension T when the radius of the drop is R. When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper.

14. If the radius of the opening of the dropper is r, the vertical force due to the surface tension on the drop of radius R (assuming r <<R) is

(A) $2\pi rT$ (B) $2\pi RT$ (C) $\frac{2\pi r^2 T}{R}$ (D) $\frac{2\pi R^2 T}{r}$

15. If $r = 5 \times 10^{-4}$ m, $\rho = 10^3$ kgm^{-3}, $g = 10$ ms^{-2}, $T = 0.11$ Nm^{-1}, the radius of the drop when it detaches from the dropper is approximately

(A) 1.4×10^{-3} m (B) 3.3×10^{-3} m (C) 2.0×10^{-3} m (D) 4.1×10^{-3} m

16. After the drop detaches, its surface energy is

(A) 1.4×10^{-6} J (B) 2.7×10^{-6} J (C) 5.4×10^{-6} J (D) 8.1×10^{-6} J

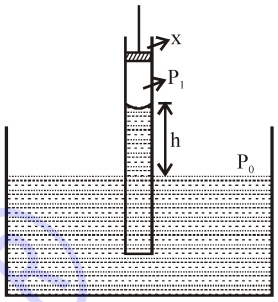
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SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**

(upto second decimal place)

17. A capillary of radius r = 0.2 mm is dipped vertically in a liquid of density, (ρ = 1gm / cc). A small piston (x) is inserted inside the capillary which maintains the constant pressure of air ($P_1 = 2 \times 10^4$ dynes/cm^2) above the hemispherical meniscus. If ambient pressure is $P_0 = 10^5$ dynes/cm^2 and surface tension of liquid is T = 72 dynes/cm, then find the height h of liquid in cm. (weight of curved part of liquid is ignored)



SECTION-III

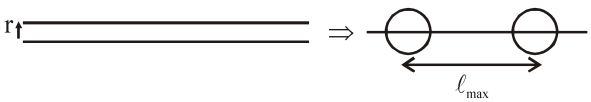
Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

18. A glass rod of diameter $d_1 = 1.5$ mm is inserted symmetrically into a glass capillary with inside diameter $d_2 = 2.0$ mm. Then the whole arrangement is vertically oriented and brought in contact with the surface of water. To what height (in cm) will the water rise in the capillary? Take surface tension of water 75 milli N/m.

19. A glass capillary sealed at the upper end is of length 0.11 m and internal diameter 2×10^{-5} m. This tube is immersed vertically into a liquid of surface tension 5.06×10^{-2} N/m. When the length $x \times 10^{-2}$ m of the tube is immersed in liquid then the liquid level inside and outside the capillary tube becomes the same, then the value of x is : (Assume atmospheric pressure is $1.01 \times 10^5 \frac{N}{m^2}$)

20. Due to the excess pressure of surface tension find the change in the radius (in Å) of a liquid drop of radius 1 mm, surface tension 0.075 N/m and Bulk modulus $1.25 \times 10^{+8}$ N/m^2.

21. On long strands form by cobwebs, water vapor condense at night to form long cylinders of water. The cylinders may collapse to form spherical droplets separated by a certain distance. Find the maximum distance (in mm) between droplets formed. Surface tension = α, radius of cylinder r = 2 mm.



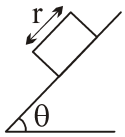
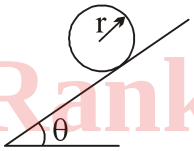
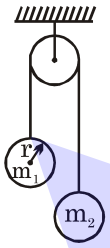
22. A square plate of 1m side moves parallel to a second plate with velocity 4 m/s. A thin layer of water exist between plates. If the viscous force is 2 N and the coeff of viscosity is 0.01 poise then the distance between the plates in mm is.

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SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

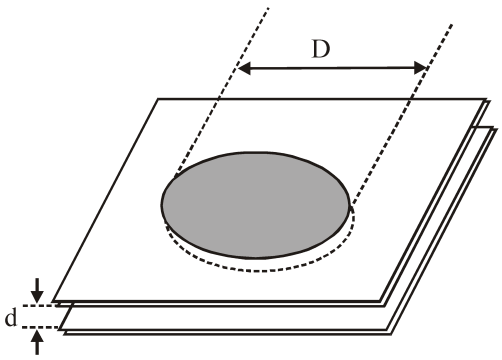
23.

Column I	Column II
(A) Body is a block lying on a rough inclined plane. Angle of inclination θ < angle of repose. A is inclined surface 	(P) Force exerted by A on body can be equal to it's weight in magnitude.
(B) Body is a ball falling in liquid with terminal velocity. A is liquid. 	(Q) If density of body is somehow increased without changing r, it's state of motion (constant acceleration / constant velocity) or equilibrium position will be disturbed.
(C) Body is a spherical ball rolling purely on inclined plane; inclined at an angle equal to angle of repose. The inclined surface is A. 	(R) Work done by the force exerted by A on the body is zero.
(D) Body is m_1 hanging on an ideal pulley system. A is string $m_1 < m_2$. 	(S) Mechanical energy of body is constant as time passes.
	(T) The kinetic energy of body is constant as time passes

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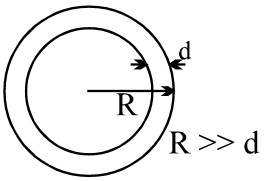
Subjective Type **5 Q. [4 M (0)]**

24. Water, which wets glass, is stuck between two parallel glass plates. The distance between the plates is d, and the diameter of the trapped water 'disc' is D >> d. What is the force acting between the plates?

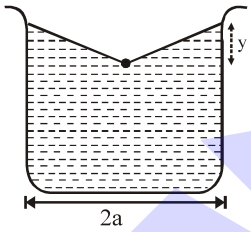


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25. A soap bubble has radius R and thickness d (<< R) as shown. It collapses into a spherical drop. Find the ratio of excess pressure in the drop to the excess pressure inside the bubble.



26. A container of width 2a is filled with a liquid. A thin wire of weight per unit length λ is gently placed over the liquid surface in the middle of the surface as shown in the figure. As a result, the liquid surface is depressed by a distance y (y << a). Determine the surface tension of the liquid. **[2004, 2M]**



27. A long capillary of radius r is immersed in liquid of surface tension T. The tube is slowly pulled out of the liquid vertically. Find the length of liquid column left in the capillary tube when it is out of liquid. (ρ density of liquid) g acceleration due to gravity. Assume liquid wets the capillary completely.

28. The velocity distribution in a viscous flow over a plate is given by $u = 4y - y^2$ for $y \le 2m$ where u is velocity in m/s at a point which is at a distance y from the plate. If the coefficient of dynamic viscosity is 1.5 in SI units. Determine the shear stress at y = 0 in N/m^2.

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SECTION-I

Single Correct Answer Type **7 Q. [3 M (–1)]**

1. Ans. (C) 2. Ans. (B) 3. Ans. (A) 4. Ans. (A) 5. Ans. (B) 6. Ans. (C)
7. Ans. (A)

Multiple Correct Answer Type **6 Q. [4 M (–1)]**

8. Ans. (A,C) 9. Ans. (A, D) 10. Ans. (B, D) 11. Ans. (A,B,D) 12. Ans. (A,B,C)
13. Ans. (B,C)

Linked Comprehension Type (1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

14. Ans. (C) 15. Ans. (A) 16. Ans. (B)

SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**
(upto second decimal place)

17. Ans. 80.72

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

18. Ans. 6 19. Ans. 1 20. Ans. 4 21. Ans. 9 22. Ans. 2

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

23. Ans. (A) P, R,S, T ; (B) P, Q, T ; (C) R, S (D) Q

Subjective Type **5 Q. [4 M (0)]**

24. Ans. $\frac{\pi D^2 Ys}{2d}$ 25. Ans. $\left(\frac{R}{24d}\right)^{\frac{1}{3}}$ 26. Ans. $\frac{\lambda a}{2y}g$ 27. Ans. $h = \frac{4T}{r\rho g}$ 28. Ans. 6

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